Paper 53: The VCML Behavioral Phase Diagram

Five Axes of Regime Variation in a Viability-Constrained Coupled Map Lattice

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Abstract

Papers 0–52 of the VCSM series each produced its predicted result plus at least one surprise. This paper argues that the surprises are not evidence of poor understanding but of a generative rather than descriptive system: simple local rules producing a behavioral space substantially larger than intuition predicts. We synthesize the experimental evidence into a unified behavioral phase diagram for VCML, identifying five experimentally separable control directions, their regime boundaries, and the mechanism responsible for each transition. Axis I (perturbation intensity, coll/site) is non-monotone with a peak at coll/site ≈ 0.004. Axis II (spatial resolution, $\delta = W_{\text{zone}}/r_{\text{wave}}$) has a hard threshold at $\delta \approx 2.5$ below which identity encoding fails. Axis III (formation flux, $\Omega = \text{WR} \times \phi_w$) is the operative causal coordinate for perturbation rate. Axis IV (write policy) is two-dimensional: trigger selectivity independently controls adversarial resistance while content determines identity encoding strength; removing the sign of the contrast signal (taking |mid|) is the only ablation to produce parity failure (Paper 52). Axis V (scale) reveals a finite correlation length: above ∼25,600 sites per zone, the copy-forward loop can no longer maintain zone-wide gradients and structure forms in patches. Together these five axes define a phase space that explains why each experiment generated structural surprise: results were correct but incomplete slices of a higher-dimensional object. The VCML region is a minimal causal field system whose behavioral phase diagram contains at least five distinct regime transitions, each mechanistically interpretable.

1 Introduction

After 52 papers, a pattern is clear. Each experiment tested a specific hypothesis and confirmed it – but also produced behavior that was not predicted by the mental model built up to that point. Paper 52 alone revealed that the write policy space is two-dimensional (trigger selectivity × write content), that removing the sign of the contrast signal is the only condition to break identity encoding, that C_perturb achieves better adversarial resistance than C_near despite 83× fewer writes, and that C_raw encodes zone identity more strongly than standard signed mid_mem. None of these findings contradicted the theory. All of them extended it in a direction the theory had not fully specified.

There are two possible interpretations of a system that keeps surprising:

1. the system is chaotic and poorly understood; or
2. the system is *generative* — small rule sets producing a behavioral space that substantially exceeds the capacity of any finite mental model.

The evidence strongly favors interpretation 2. The surprises in Papers 0–52 are not random. They are structurally located: each one corresponds to a new axis of the behavioral phase space becoming visible. This paper maps those axes.

The generative vs. descriptive distinction. A descriptive system encodes its input. A generative system produces structure that was not explicit in its rules. The distinction matters for prediction: a descriptive system is exhausted by a faithful description of its rules, while a generative system has behavioral surface that exceeds its rule complexity. VCML is generative at multiple levels: the copy-forward loop generates spatial structure without spatial rules; the viability gate generates temporal discrimination without temporal rules; the three-timescale consolidation generates long-term memory without recurrence. The result is a system with a behavioral phase diagram whose structure cannot be read directly off the rule description.

2 Five Experimentally Separable Control Directions

2.1 Axis I: Perturbation Intensity (coll/site)

Source: Papers 87/88 (amplitude sweep), V93 (WR sweep at S3).

Finding: sg4 is non-monotone in perturbation intensity. At optimal coll/site ≈ 0.004, sg4 reaches its peak value (0.038 at S3 with 2× WR). Below this threshold, insufficient collapse-birth events occur for the copy-forward loop to refresh spatial structure. Above it, each site is re-collapsed before consolidation can complete, erasing mid_mem before fieldM updates.

Regime boundaries:

- coll/site < 0.003: sub-optimal zone (formation starved)
- coll/site $\in [0.003, 0.006]$: adaptive peak (optimal)
- coll/site > 0.010: over-perturbed zone (consolidation disrupted)

Mechanism: The copy-forward loop requires collapse events to propagate fieldM; too few events starve propagation. The quiescence gate requires calm streaks $\geq$ SS; too many collapses prevent the streak from accumulating. The peak is the point where both constraints are simultaneously satisfied.

Dynamic advantage: The dynamic condition (C_ref) consistently outperforms static at the adaptive peak. At WR_MULT=2× (S3), dynamic advantage is +829%. Over-perturbation eliminates the advantage and eventually inverts it (V93, WR_MULT=10×, dynamic advantage −29%).

2.2 Axis II: Spatial Resolution ($\delta = W_{\mathbf{zone}}/r_{\mathbf{wave}}$)

Source: Paper 47 (C2 coordinate validation, delta sweep).

Finding: na_ratio peaks at $\delta \approx 5$ and degrades toward unity at $\delta \lesssim 2.5$, below which waves launched in one zone substantially overlap adjacent zones (wave bleed). At $\delta = 1.25$, na_ratio drops below 1 (parity territory).

Regime boundaries:

- $\delta < 2.5$: wave-bleed regime (parity encoding or no structure)
- $\delta \geq 2.5$: identity-encoding regime
- $\delta \geq 5$: plateau (additional spatial resolution provides diminishing returns)

Mechanism: When δ is small, a wave initiated in zone A collapses cells in zone B before decaying. The copy-forward loop propagates zone A fieldM into zone B sites, erasing the boundary. δ must be large enough that the wave is zone-contained. This is a hard threshold, not a soft degradation.

C2 consequence: δ is the third coordinate of the C2 causal triplet (Ω, SS, δ). It sets the spatial resolution of the memory system independently of the perturbation rate.

2.3 Axis III: Formation Flux ($\Omega = \mathrm{WR} \times \phi_w$)

Source: Papers 46–47.

Finding: $\Omega = \mathrm{WR} \times \phi_w$ (wave rate times wave occupancy fraction) is the operative coordinate for perturbation rate. Sweeping WR at constant Ω produces approximately flat sg4 (CV=0.126, Paper 46). Sweeping WR at constant zone width produces non-flat results because changing WR also changes Ω.

Regime: This axis does not have a simple phase boundary; it scales approximately linearly with formation rate until coll/site exceeds the Axis I over-perturbation threshold. Ω and coll/site are coupled: increasing WR increases both.

2.4 Direction IV: Write Policy Space (Trigger × Content)

Source: Paper 52 (boundary-coupled distinction).

Note on status: Unlike Axes I–III and V, which are continuous physical parameters with clear order parameters and threshold transitions, the write policy direction is an algorithmic policy space: a set of design choices about when and what to write, not a single continuous control variable. We include it as a control direction rather than a strict thermodynamic axis because it produces two behaviorally distinct subspaces (trigger selectivity and write content) that are experimentally orthogonal and mechanistically interpretable.

Finding: The write policy space is two-dimensional.

Content axis. What the cell writes to fieldM:

- **C_raw** (raw hid): na_form = 1.802 — strongest identity encoding. The raw hidden state carries more zone-discriminative signal than mid_mem.
- **Signed mid** (standard VCSM): na_form $\in [1.1, 1.3]$ — subtracting baseline_h reduces identity signal while adding boundary sensitivity.
- **|mid|** (C_abs): na_form = 0.877 — the only parity failure among 9 conditions. Removing the sign of the contrast signal destroys the directional information that makes intergenerational propagation zone-informative.

The sign is load-bearing: removing it destroys the directional contrast information necessary for zone-discriminative intergenerational propagation. Why zones require the sign is mechanistically plausible (suppressed zones push viability downward, excited zones push it upward, so the signed direction distinguishes them), but the precise causal pathway is a target for future experiment. Destroying the sign removes whatever zone-discriminating information it carries.

Trigger axis. When the cell writes to fieldM:

- **C_perturb**: writes only during active wave events (writes-per-site = 0.0004). Cosine maintenance = 0.664 — best adversarial resistance.
- **C_near**: writes when near viability boundary. wps = 0.033. cos_maint = 0.546.
- **C_ref**: writes whenever quiescent (wps = 0.607). cos_maint = −0.274 — re-encodes adversarial environment.
- **C_far**: writes far from boundary (wps = 0.476). cos_maint = −0.085 — also re-encodes.

High write frequency = rapid adaptation = inability to distinguish formation from adversarial environment. Low write frequency = structural commitment = adversarial resistance.

Design tradeoff. The two axes optimize different objectives. Moving along the content axis (toward C_raw) increases zone identity encoding strength at the cost of losing boundary sensitivity. Moving along the trigger axis (toward C_perturb) increases adversarial resistance at the cost of slower formation. Standard VCSM (C_ref) occupies the responsive corner: high frequency, signed content — rapid formation, rapid re-encoding.

C_rand finding. Random trigger (matched write rate, wps = 0.001) achieves na_form = 1.335 — highest among standard-content conditions. This confirms that boundary proximity is one valid trigger criterion among many; stochastic subsampling is sufficient for formation.

2.5 Axis V: Scale and the Finite Correlation Length

Source: V94 (scale-invariant metric sweep).

Finding: sg4_norm (normalized by within-zone variance) is not flat across grid sizes. S1 (0.4270) → S2 peak (0.4593) → S3 (0.3647) → S4 (0.1874). At S4 (102,400 active sites, zone width ~ 320 sites), na_ratio inverts (0.732, parity territory).

Mechanism: The copy-forward loop is local (wave radius ≈ 2 sites). At S1–S3, the loop can propagate fieldM across the zone width within one wave episode. At S4, zone width exceeds the loop's correlation length. Structure forms in patches of size $\sim r_{\text{wave}}$, not in zone-wide gradients. Adjacent patches may encode different patterns, degrading the na_ratio (adjacent zones may look more similar than non-adjacent when both form patches at random positions).

Regime boundary: Approximately $N_{\text{sites/zone}} \sim 25\,000$ (S3 edge). Below this, zone-wide gradients form. Above this, patch formation dominates.

Implication: Geographic or hierarchical coupling is required to maintain structure at scales beyond the correlation length. A regional module (small, short correlation length) could encode summaries of the large-zone state, while the large zone encodes fine-grained spatial detail. This is the architectural argument for multi-region VCSM.

3 Cross-Axis Dependencies and Interactions

The five control directions are experimentally separable but not causally independent. Three coupling mechanisms produce the most important cross-axis interactions.

Axes I and III are coupled through WR. Increasing WR simultaneously increases Ω (Axis III) and coll/site (Axis I). This means the optimal WR is not a single number but a function of grid size: the Axis I peak (coll/site ≈ 0.004) corresponds to different WR values at different scales. V93 shows the optimum is WR $\approx 2\times$ the base rate at S3; V92 shows the optimum shifts at S1.

Axes II and V interact through zone width. $\delta = W_{\text{zone}}/r_{\text{wave}}$. Increasing N_{active} (Axis V) while holding r_{wave} fixed also increases δ. S4 has $\delta \approx 160$ — well above the Axis II threshold. The S4 patch formation failure is therefore not an Axis II failure (insufficient δ); it is a correlation-length failure that occurs even when δ is large. These are distinct mechanisms.

Axis IV interacts with Axis I at the C_perturb extreme. C_perturb writes only during active wave events. In the Axis I sub-optimal regime (very low coll/site), wave events are rare and brief. C_perturb would then write essentially nothing, preventing formation entirely. The trigger selectivity axis is only meaningful when the system is in the Axis I adaptive zone. This is a testable cross-axis prediction: C_perturb should fail at low coll/site while C_ref continues to form structure (slowly).

4 The VCML Behavioral Phase Diagram

Figure 1 presents the phase diagram across four of the five axes.

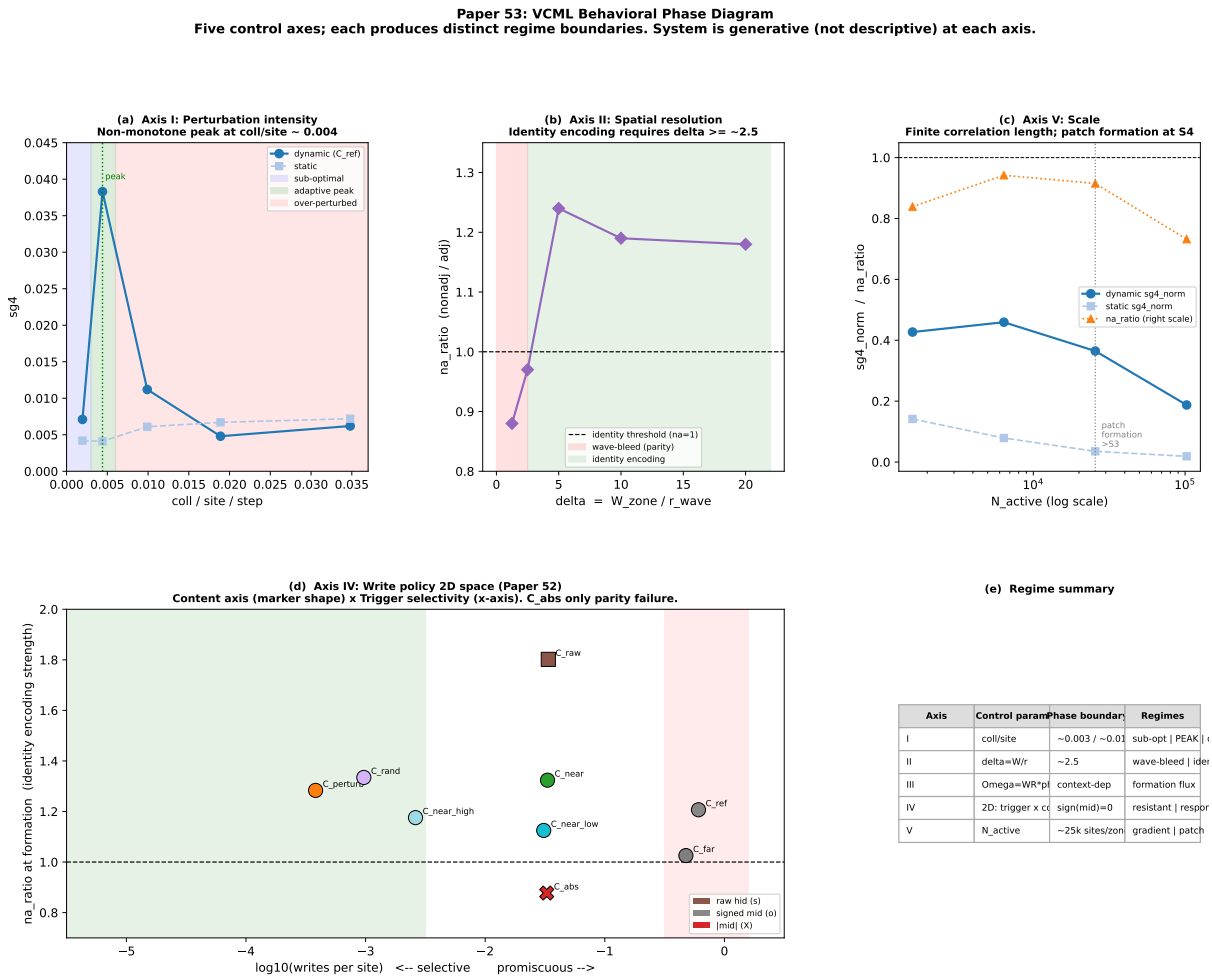

Figure 1: **VCML behavioral phase diagram.** (a) Axis I: perturbation intensity. Non-monotone peak at coll/site ≈ 0.004; shaded regions indicate sub-optimal, adaptive, and over-perturbed regimes. (b) Axis II: spatial resolution. Identity encoding requires $\delta \geq 2.5$; hard threshold from wave bleed. (c) Axis V: scale. sg4_norm (normalized structural signal) declines beyond S3 ($N \approx 25\,000$ sites/zone); na_ratio inverts at S4. Finite correlation length of the copy-forward loop. (d) Axis IV: write policy 2D space. Each Paper 52 condition placed at ($\log_{10}$(wps), na_form); marker shape encodes content type (raw hid = square, signed mid = circle, |mid| = cross). C_abs (only parity failure) lies at the intersection of boundary-trigger selectivity and unsigned content. (e) Regime summary table.

4.1 VCML as a Minimal Causal Field System

A minimal causal field system is a system in which the same local rules produce, depending on the control-parameter regime:

1. identity encoding (nonadj/adj > 1, zone-discriminative fieldM);
2. adversarial resistance (fieldM stable under input reversal);
3. intergenerational memory (structure maintained through complete cell turnover);

4. spatial topology formation (zone-specific gradients without spatial rules).

VCML satisfies all four, and the phase diagram shows these are not always simultaneously active: the same system in the over-perturbed regime loses identity encoding while retaining turnover; in the C_abs regime, it retains adversarial stability while losing identity encoding; at S4, it retains formation while losing zone-wide coherence.

This is precisely the hallmark of a phase diagram: the same system in different regimes. The claim "VCML is a minimal causal field system" is the claim that the behavioral phase diagram is non-trivial (multiple distinct regimes) and mechanistically interpretable (each transition has a local causal explanation).

4.2 Why the System Kept Surprising

Each of Papers 0–52 explored a one-dimensional slice of the five-dimensional phase space without knowing the full dimensionality. The surprises were the system crossing a regime boundary along a dimension the current mental model had not separated.

The relay papers (35–50) failed to predict parity encoding because the na_ratio axis (Axis II/δ regime) had not been identified. The amplitude papers (87–88) produced non-monotone behavior because Axis I's over-perturbation regime was not in the mental model. Paper 52 produced C_perturb as the best adversarial condition because the trigger selectivity axis had not been separated from the content axis.

Once the axes are separated, each result becomes a predictable location in the phase diagram rather than a surprise.

5 Testable Predictions

The phase diagram generates three cross-axis predictions not yet tested.

1. **C_perturb failure at low coll/site.** C_perturb writes only during active wave events. At Axis I sub-optimal (coll/site < 0.002), waves are rare. Prediction: C_perturb formation fails (na_form < 1) while C_ref maintains formation (na_form > 1, slowly). This would confirm that the trigger axis is contingent on the perturbation intensity axis.

2. **Scale recovery via hierarchical coupling.** At S4 (N=102,400), patch formation dominates because the copy-forward loop correlation length is shorter than zone width. A hierarchical coupling — a small upper-layer VCML encoding zone-level fieldM summaries — should restore zone-wide coherence even at S4. Prediction: na_ratio at S4 recovers to > 1 under hierarchical coupling.

3. **C_raw adversarial performance.** C_raw achieves highest na_form (1.802) but its adversarial maintenance (cos_maint = 0.777) was measured only at standard coll/site. Prediction: C_raw adversarial resistance should be higher than C_near at low coll/site (where the raw hid signal is more stable) but lower than C_near at high coll/site (where raw hid is noisier due to rapid collapse cycles).

6 Discussion

6.1 The Logistic Map Analogy

The logistic map $x_{t+1} = rx_t(1 - x_t)$ generates fixed points, oscillations, and chaos from a single parameter. VCML generates identity encoding, adversarial resistance, intergenerational memory, and patch formation from five interacting parameters. The structural parallel is analogical, not equivalential: VCML is not a one-dimensional map and does not exhibit period-doubling in

the standard sense. The analogy holds in structure, not detail. Both systems have threshold transitions (the logistic chaos onset; VCML's δ threshold and amplitude peak), both produce non-monotone behavior in a control parameter, and both generate behavioral structure that was not anticipated from the rule description. Small rule sets at a generative layer produce behavioral diversity exceeding the intuition of anyone studying only the rules.

6.2 Implication for the A≠B Attack

A persistent reviewer objection is that VCSM's contrast signal $\Delta = \text{hid} - \text{baseline_h}$ is merely subtraction, already present in predictive coding. The phase diagram answers this at the axis level. Predictive coding operates entirely on the content axis of Axis IV. The trigger axis (when subtraction results are written to long-term memory, gated by survival and quiescence), the Axis I perturbation intensity regime (requiring collapse-birth cycles, not just prediction error), and the Axis II spatial resolution requirement are structural elements absent from predictive coding. The subtraction is not the contribution; the phase diagram location is.

6.3 Implication for Scale

The finite correlation length (Axis V) is not a failure of VCSM. It is a prediction. Any memory system whose consolidation mechanism is local will have a correlation length. The VCML correlation length is approximately the wave radius (~ 2 sites). Systems with local learning rules and no global readout (e.g., cortical columns) should show analogous patch formation above a critical zone size. The prediction is falsifiable: increasing the wave radius should proportionally increase the scale at which patch formation begins.

6.4 Open Question: Is There a Unified Order Parameter?

The five control directions presented here may not be the fundamental coordinates of the system. Paper 48 introduced sg_C, the causal structure metric: $\text{sg_C} = \text{sg4}/\sigma_w$ (structural signal divided by within-zone noise). sg_C was motivated by the observation that sg4 and σ_w move in opposite directions under certain interventions, making the ratio the true quality measure.

A natural question is whether sg_C functions as a unified order parameter that collapses the five-axis regime diagram. If sg_C is the right order parameter, each axis might correspond to a different way of varying the signal/noise ratio rather than to genuinely distinct physical regimes. Under this interpretation, Axis I (amplitude) controls the signal amplitude; Axis II (δ) controls zone bleed into noise; Axis IV (write policy) controls write selectivity (signal-gating); Axis V (scale) controls the noise floor via patch averaging. If all five axes reduce to variations of sg_C, the regime diagram would collapse to a single one-dimensional object — which would make the theory considerably stronger, not weaker.

This remains an open question. The current paper takes the empirical approach: five separable control directions are identified from experimental data, each with a mechanistic interpretation. Whether they reduce to a common order parameter is the natural target for Paper 54.

7 Conclusion

Papers 0–52 were each exploring a one-dimensional slice of a multi-dimensional control space. The surprises were regime crossings along unmapped directions. This paper identifies five experimentally separable control directions, their boundaries, and their mechanistic interpretation. Axis IV is explicitly distinguished as a policy space rather than a continuous thermodynamic parameter. The result is a behavioral phase diagram for VCML that explains retroactively why each result was correct but incomplete, and generates three testable cross-axis predictions.

An open question — whether the five directions reduce to a single order parameter (sg_C) — is identified as the target for Paper 54. If they do, the diagram becomes more fundamental, not less.

VCML is a minimal causal field system: a small rule set sitting at a generative layer, producing behavioral diversity that cannot be read off the rule description. The correct question is not whether the mechanism is understood, but which regime of the control space a given experiment inhabits.